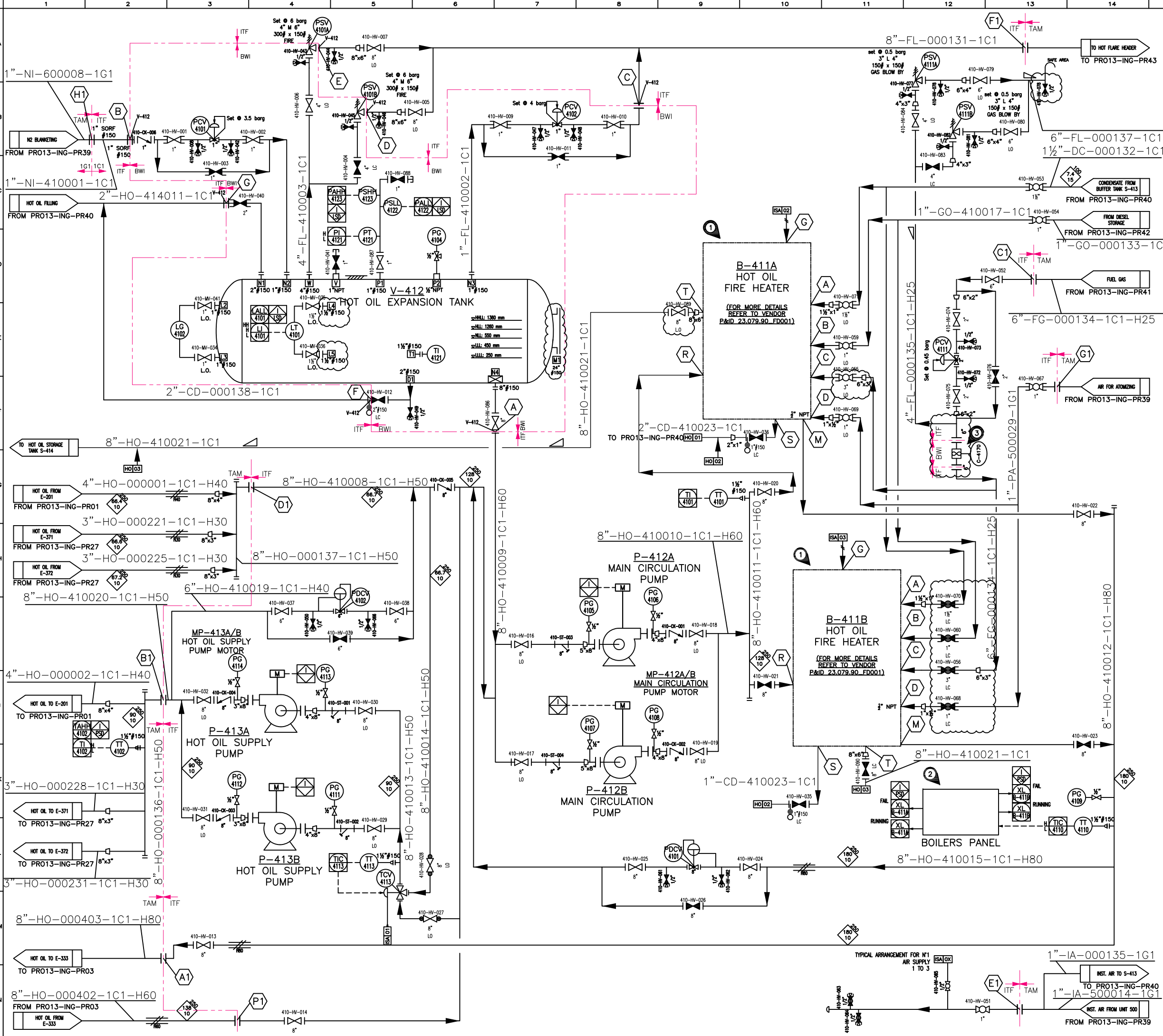




REFERENCE DOCUMENTS									
P&ID LEGEND					PRO13-ING-PROO 21044-PR-PID-00000				
PROCESS FLOW DIAGRAM					PRO10-ING-PRO1/02/03/04 21044-PR-PFD-00001				
UTILITY FLOW DIAGRAM					PRO10-ING-PRO5/11 21044-PR-UFD-00001				
MATERIAL SELECTION DIAGRAM					PRO08-ING-FR01 21044-PR-MSD-00001				
V-412 HOT OIL EXPANSION TANK DESIGN PRESSURE 8 bar g DESIGN TEMPERATURE -15/300 °C INTERNAL DIAMETER 1800 mm TL TO TL 4350 mm CORROSION ALLOWANCE 1.5 mm INSULATION H(80) mm LINING/CLADDING N/A MATERIAL P275 NH(or EQ) INTERNAL VOLUME 12.6 m3 INTERNAL ELEMENTS -					P-412A/B MAIN CIRCULATION PUMPS SPEED 2974 rpm RATED CAPACITY 240 m3/h HEAD 2.6 bar TYPE CENTRIFUGAL MANUFACTURER KSB CASING DESIGN PRESSURE 15 barg CASING DESIGN TEMPERATURE -15/250 °C MODEL ETW 125 MOTOR POWER 45 kW CASING MATERIAL CAST IRON				
P-413A/B HOT OIL SUPPLY PUMPS SPEED 2977 rpm RATED CAPACITY 105 m3/h HEAD 3.1 bar TYPE CENTRIFUGAL MANUFACTURER KSB CASING DESIGN PRESSURE 16 barg CASING DESIGN TEMPERATURE -15/250 °C MODEL ETW 100 MOTOR POWER 30 kW CASING MATERIAL CAST IRON					MP-412A/B MAIN CIRCULATION PUMP ELECTRIC MOTOR SPEED 2974 rpm RATED POWER 45 kW VOLTAGE 690 Vac FREQUENCY 50 Hz STARTING SOFT START ENCLOSURE PROTECTION IP55 STD IEC FRAME SIZE 225M				
MP-413A/B HOT OIL SUPPLY PUMP ELECTRIC MOTOR SPEED 2977 rpm RATED POWER 30 kW VOLTAGE 690 Vac FREQUENCY 50 Hz STARTING SOFT START ENCLOSURE PROTECTION IP55 STD IEC FRAME SIZE 200 L					B-411A/B HOT OIL FIRE HEATER DESIGN DUTY 5800 kW DESIGN PRESS. SHELL 0.1 bar g DESIGN TEMP. SHELL -15/450 °C DESIGN PRESS. TUBE 10 bar g DESIGN TEMP. TUBE -15/300 °C SHELL INTERNAL DIAMETER 2480 mm COIL TOT. LENGTH 747 mm CORR. ALL. SHELL/TUBE 0/0 mm INSULATION AIR-COOLED NULL MATERIAL SHELL SIDE CS MATERIAL TUBE SIDE P235 GH SURFACE AREA 192 m2				
ISA 04 INSTRUMENT SUPPLY AIR									
HOT 04 HOT OIL									

MP-413A/B

HOT OIL SUPPLY PUMP ELECTRIC MOTOR

SPEED

RATED POWER

VOLTAGE

FREQUENCY

STARTING

ENCLOSURE PROTECTION

STD

FRAME SIZE

2977 rpm

30 kW

690 Vac

50 Hz

SOFT START

IP55

EC

200 L

B-411A/B

HOT OIL FIRE HEATER

DESIGN DUTY

DESIGN PRES. SHELL

DESIGN TEMP. SHELL

DESIGN PRES. TUBE

DESIGN TEMP. TUBE

SHELL INTERNAL DIAMETER

COIL TOT. LENGTH

INSULATION

CORR. ALL SHELL/TUBE

MATERIAL SHELL SIDE

MATERIAL TUBE SIDE

SURFACE AREA

5800 kW

0.1 bar g

-15/450 °C

10 bar g

-15/300 °C

2480 mm

747 mm

0/0 mm

AR-COOLED WALL

CS

P235 GH

192 m2

ISA 01

INSTRUMENT SUPPLY AIR

HO 01

HOT OIL